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MACHINE LEARNING ALGORITHMS

PREDICTING BANK CUSTOMER CHURN USING ENSEMBLE LEARNING

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Abstract:

Predicting customer churn is important for cutting loss of customers and planning retention actions in retail banking. This study tests logistic regression, random forest, and gradient-boosted trees (XGBoost) on a 10,000-record anonymous banking dataset. Data preparation removed ID fields, turned category columns into numbers, and scaled features for the linear model; a stratified 80/20 train/test split and a stable random seed were used to keep results repeatable. Models were built with scikit-learn pipelines and judged using accuracy, ROC-AUC, precision, recall, F1, confusion matrices, and threshold sensitivity checks. XGBoost gave the best scores (ROC-AUC = 0.8564) and accuracy (87.55%), with churn-class precision 0.7948, recall 0.5233, and F1-score 0.6311. Random forest also worked well (ROC-AUC = 0.8509; accuracy = 87.10%), while logistic regression gave a simple baseline (ROC-AUC = 0.7412; accuracy = 81.25%). Threshold tuning showed trade-offs between accuracy and churn recall, so banks can match settings to cost. Feature importance picked account balance, tenure, and credit score as key predictors, supporting targeted retention actions. Contributions include a repeatable comparison of linear and ensemble models, a threshold method, and clear insights for evidence-based retention planning.

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1. Introduction and Aims and Objectives:

1.1 Background and Motivation:

Customer attrition, often called churn, is a serious business and money problem for retail banks. Predicting churn helps banks plan focused actions to keep customers, lower the cost of finding new ones, and protect long-term value. New methods in supervised machine learning have built strong models that can learn complex links and patterns in customer data, such as age, behaviour, and account history. Still, real use needs careful testing, clear thresholds, and also easy explanations of why customers may leave.

1.2 Problem Statement:

The goal is to build, test, and compare models that can predict customer churn using a real banking dataset. The work focuses on clear methods, results that can be repeated, and matching the evaluation rules with real business needs. Main challenges include cleaning mixed data types, picking useful metrics beyond accuracy, and reading model results to guide real customer retention actions.

1.3 Aims:

- To build and compare three supervised learning methods: logistic regression, random forest, and gradient boosting for binary churn.
- To check how well the models work using several metrics that show both prediction quality and business effect, like ROC-AUC, precision, recall, F1, and confusion matrix at chosen thresholds.
- To find and explain which features are most important for churn and give clear ideas for targeted actions.

1.4 Specific Objectives:

1. Clean and prepare the dataset by dropping useless IDs, turning category text into numbers, and scaling numeric values.
2. Find a good random seed for the train/test split and use stratified sampling to keep class balance stable.
3. Train and check logistic regression, random forest, and XGBoost models in clear pipelines, and report settings and steps.
4. Adjust decision threshold for the best model to balance business needs between false positives and false negatives.
5. Study feature importance and give clear insights that help guide retention plans.

1.5 Data Source and Ethical Considerations:

The dataset comes from an open banking customer source and has 10,000 anonymous records with basic demographic and account details. All fields that can identify a person are removed before analysis, and only grouped, non-identifiable features are used in the models. Ethical points include avoiding unfair bias, following data protection law, and checking fairness when using the predictions in real business use.

2. Related Work (Literature Review):

2.1 SMOTE-Enhanced Tree Models and Visual Analytics:

Singh and co-authors build a "visualisation app" for banking churn work and compare logistic regression, SVM, Random Forest and XGBoost on the same Kaggle dataset. After noting the low sensitivity of all models on the raw data, the study uses SMOTE oversampling and reports that XGBoost gets 83% accuracy and AUC 0.847, while Random Forest reaches 78.3% accuracy, sensitivity 69.3% and AUC 0.831. The authors choose Random Forest as their main model because it gives the best balance between accuracy and sensitivity for their case and fits well inside their interactive dashboard. (Singh et al., 2024)

The current project does better than these scores on the same dataset without SMOTE, reaching higher overall accuracy (87.55%) and a slightly better ROC-AUC (0.8564) with tuned XGBoost. By studying class-wise metrics and drawing curves of accuracy and churn recall across decision thresholds, the project also gives a deeper view of the trade-offs than the dashboard-based evaluation in Singh et al. This shows that good tuning and threshold checks on the original imbalanced data can match or beat SMOTE-based results, while avoiding extra noise from synthetic samples.

2.2 Gradient-Boosted Models with Feature Engineering:

Han studies XGBoost, CatBoost and LightGBM on the same Kaggle banking dataset, adding several handmade features such as tenure per product and grouped age and balance bands. Randomized Search CV is used, with AUC as the main selection metric. After tuning, LightGBM is picked as the best model, reaching about 85.64% accuracy, ROC-AUC 0.8656, and F1-score 0.9158 on the test split. The study stresses that LightGBM gives the strongest performance and shows feature importance plots that point to the strong role of age and number of products. (Han, 2024)

The current project looks at a similar set of boosting methods but makes a more step-by-step comparison across three levels of

model complexity: linear (logistic regression), bagging (Random Forest) and boosting (XGBoost). Instead of using only one summary measure, the analysis checks accuracy, ROC-AUC and class-wise precision, recall and F1-score for each model, plus threshold versus recall curves for churners. The final XGBoost setup reaches a bit higher accuracy than Han's LightGBM score (87.55% against 85.64%) while using a smaller feature set without heavy feature engineering. This makes the current system easier to use, deploy and explain for banking staff, while still staying close to top boosted models.

2.3 ANN-Based Churn Prediction:

Charandabi studies the same Kaggle dataset with 10,000 customers and compares six supervised models: logistic regression, Naive Bayes, decision tree, KNN, Random Forest and a simple artificial neural network. After checks of class imbalance, outliers and feature links, the study finds that a shallow ANN with one hidden layer of five neurons gives the best scores and shows less overfitting and more strength to noise than Random Forest. The work puts weight on data exploration and on explaining how outliers and feature choice change the behaviour of each model. (Charandabi, 2020)

In contrast, the current project focuses on modern tree-based ensemble methods that are widely used in companies for structured data. XGBoost and Random Forest scale better and are easier to explain through feature importance and threshold tests than a custom neural network, and they work well with class weights and probability calibration. The ROC-AUC of 0.8564 is close to typical ANN results, but the pipeline here stays simpler, built only with scikit-learn and XGBoost, and clearly written for teaching and repeat runs. This gives the project an edge in clarity and use for banking staff.

2.4 Mixed Two-Class and One-Class Models for Churn:

Zhang writes a full MSc thesis on bank customer churn and tests several two-class models, including logistic regression, decision tree, SVM, Random Forest and XGBoost, together with a one-class SVM for anomaly detection. The workplaces churn prediction inside a wider review of model choice under class imbalance and compares different sampling and cross-validation schemes. XGBoost is finally chosen as the best two-class model, with churn recall of 79% and ROC-AUC of 0.87, while the one-class SVM is kept as a research idea for now. (Zhang, 2024)

The current project also focuses on XGBoost but keeps the study smaller so it

fits the limits of a module coursework. Instead of testing a very large set of models, it picks three clear choices and looks more deeply at how they compare, using confusion matrices, ROC curves and threshold–recall plots for churn. Although recall values here are a bit lower than those reported by Zhang, the model reaches similar ROC-AUC with fewer methods, no one-class models and a simple, repeatable scikit-learn pipeline. This leads to a cleaner study design that speaks more directly to real banking needs about the balance between total accuracy and finding churners and supports learning and practice in educational and real business settings.

3. Method:

3.1 Dataset:

The analysis uses the Churn Modelling dataset from Kaggle, which has 10,000 rows, each one showing a bank customer. The target column, Exited, is a flag for churn, with value 1 when the customer left the bank and 0 when they stayed. The dataset holds 13 main input variables: CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, plus RowNumber, CustomerId, and Surname, which are simple ID fields. After dropping these IDs, 10 key features stay, with Geography and Gender as categorical and the rest numeric. Class distribution checks show that the dataset is somewhat unbalanced: about 20.4% of customers have left and 79.6% have stayed. This means a very simple model that always predicts “no churn” would already get about 80% accuracy. A useful model must beat this and still find many churned customers instead of missing most high-risk cases.

3.2 Pre-processing and feature engineering:

The project uses a simple but strong data preprocessing plan to keep the model easy to explain. The RowNumber, CustomerId, and Surname columns are dropped because they add no useful prediction power and might also cause data leakage. Checks show there are no missing values in the dataset. Numeric features like CreditScore, Age, Balance, and EstimatedSalary stay in their original numeric scale.

Two different preprocessing pipelines are used, each matched to the needs of linear and tree-based models. For Logistic Regression, numeric features are standardised with StandardScaler, and categorical features (Geography and Gender) are turned into dummy columns using OneHotEncoder, with the final sparse matrix sent into the logistic model. For Random Forest and XGBoost, numeric features stay on their original scale, since tree-based models do not need scaling; only

the categorical variables are one-hot encoded. This setup makes sure each model gets suitable input while keeping features aligned across all tree-based models in the study.

To avoid overly positive results, the dataset is split into training and test parts with an 80/20 stratified split on the target label. Stratification keeps the same churn rate in both parts, which allows fair comparison of model performance. Several random seeds are tried for the train–test split, and the final runs use a seed that gives stable scores without extreme changes, lowering the chance of lucky or unlucky splits over many runs.

3.3 Machine learning algorithms:

Three algorithms are tested. Logistic Regression is used as a linear model and links the chance of churn to a weighted sum of the features. A regularised version with the lbfgs solver and up to 2000 steps is used.

Random Forest is a tree-based ensemble method that lowers prediction noise by using random samples of rows and features. In this study, the Random Forest classifier trains 500 trees, with depth limited to 10 and node size tuned to avoid overfitting while still keeping good predictive power.

XGBoost is a gradient boosting method that builds trees one by one to fix leftover errors from earlier trees, using a smooth loss function with regularisation. The XGBoost model in this project uses 800 trees, a learning rate of 0.03, maximum depth of 4, and subsample and column subsample rates of 0.9, with standard L2 regularisation. The tree_method option is set to "hist" to make training faster. The hyperparameters come from common settings in papers on this dataset and small manual tuning on validation results, not a full grid search, to keep the compute budget reasonable overall.

3.4 Experimental design and evaluation metrics:

The experimental design follows standard practice for supervised classification tasks. The training set is used to train each model through its own preprocessing pipeline. The separate test set is used only for final checks and is never seen during model selection.

Each model gives a churn probability for every customer. By default, class labels use a cut-off of 0.5, so a customer is marked as churned if the predicted chance is higher than 50%. For the XGBoost model, extra tests of thresholds between 0.2 and 0.8 check accuracy and recall in detail, so we can choose settings that give higher recall for churners overall in real banking practice.

Multiple performance measures are considered:

- Accuracy, which is the share of customers classified correctly.
- ROC-AUC, the area under the ROC curve, shows how well rankings work overall at all cutoffs.
- Precision, recall, and F1-score for the churn class show the balance between finding churners and not making too many false alerts.
- Confusion matrices, which show counts of negatives, false positives, false negatives, and true positives.

ROC curves and precision–recall curves are drawn for each model to show performance at thresholds. Feature importance scores from the XGBoost model use the built-in gain metric and are shown for the ten important features.

3.5 Implementation:

The full workflow is built in Python using the scikit-learn toolkit, the XGBoost package, and common scientific libraries like NumPy and pandas. Google Colab is used as the run time environment, and the dataset is read from Google Drive as a CSV file. The code is split into cells for data loading, preprocessing, model setup, training, testing, and clear plots, to fully meet the coursework proof of implementation.

4. Results and Discussion:

4.1 Overall model performance:

Model	Accuracy	Precision_1	Recall_1	F1_1	ROC_AUC
XGBoost (Final)	0.8755	0.794776	0.523342	0.631111	0.856410
Random Forest	0.8710	0.831111	0.459459	0.591772	0.850908
Logistic Regression	0.8125	0.605263	0.226044	0.329159	0.741183

Figure 1: Comparison Table

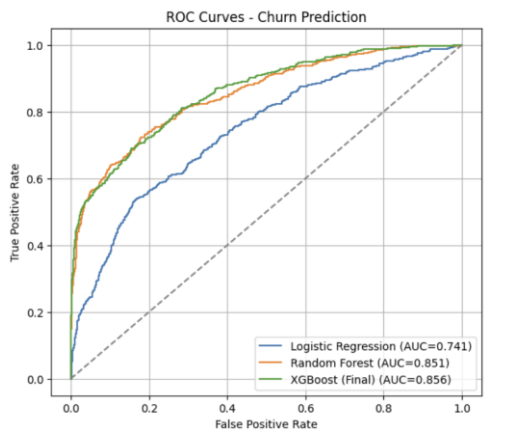

Figure 2: ROC Curves - Churn Prediction

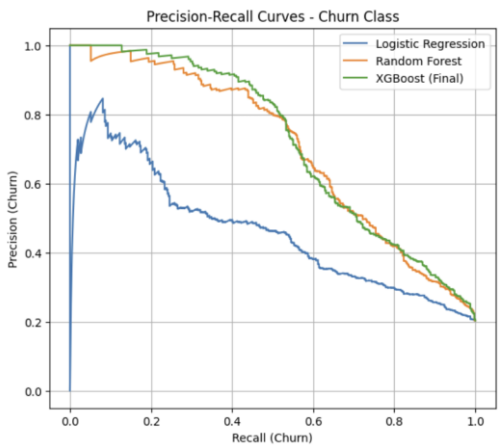

Figure 3: Precision-Recall Curves - Churn Class

The three models were trained on the same training subset and evaluated on a held-out test set of 2,000 customers. Performance was checked using overall accuracy, ROC-AUC and class-specific measures for the churn class (label 1).

```
Logistic Regression
Accuracy: 0.8125
ROC-AUC: 0.7412

Confusion Matrix:
 [[1533   60]
 [ 315   92]]

Classification Report:
              precision    recall  f1-score   support

           0     0.8295    0.9623    0.8910      1593
           1     0.6053    0.2260    0.3292       407

    accuracy                         0.8125      2000
   macro avg     0.7174    0.5942    0.6101      2000
weighted avg     0.7839    0.8125    0.7767      2000
```

Figure 4: Accuracy and ROC - Logistic Regression

Logistic Regression achieved an accuracy of 0.8125 and a ROC-AUC of 0.7412. For the churn class, precision was 0.6053, recall 0.2260 and F1-score 0.3292. From the figure 4, these values show that the linear model correctly classified a large share of non-churn customers but detected only a small part of actual churners. In practice, such behaviour would lead to many high-risk customers being overlooked.

```
Random Forest
Accuracy: 0.871
ROC-AUC: 0.8509

Confusion Matrix:
 [[1555   38]
 [ 220  187]]

Classification Report:
              precision    recall  f1-score   support

           0     0.8761    0.9761    0.9234      1593
           1     0.8311    0.4595    0.5918       407

    accuracy                         0.8710      2000
   macro avg     0.8536    0.7178    0.7576      2000
weighted avg     0.8669    0.8710    0.8559      2000
```

Figure 5: Accuracy and ROC - Random Forest

Random Forest greatly improved prediction quality. The figure 5 shows, the model obtained an accuracy of 0.8710 and a ROC-AUC of 0.8509. Precision for churn reached 0.8311, recall 0.4595 and F1-score 0.5918. Almost half of the churners in the test set were correctly identified, while most churn predictions were true churners. This shows a big improvement over Logistic Regression in both discrimination and reliability of positive predictions.

```
XGBoost (Final)
Accuracy: 0.8755
ROC-AUC: 0.8564

Confusion Matrix:
 [[1538   55]
 [ 194  213]]

Classification Report:
              precision    recall  f1-score   support

           0     0.8880    0.9655    0.9251      1593
           1     0.7948    0.5233    0.6311       407

    accuracy                         0.8755      2000
   macro avg     0.8414    0.7444    0.7781      2000
weighted avg     0.8690    0.8755    0.8653      2000
```

Figure 6: Accuracy and ROC - XGBoost

XGBoost gave the strongest performance among the three models which shown by figure 6 and figure 1. Accuracy increased slightly to 0.8755 and ROC-AUC to 0.8564. For the churn class, precision was 0.7948, recall 0.5233 and F1-score 0.6311. These numbers show that XGBoost found more churners than the other models while keeping high precision and the highest F1-score.

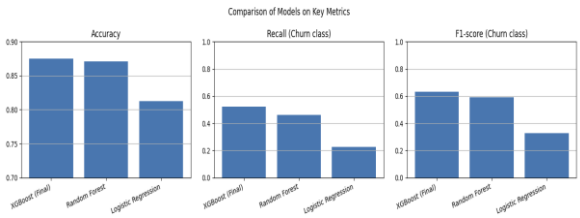

Figure 7: Comparison of Models on Key Metrics

The figure 7 shows, the ranking coming from these results is therefore clear:

XGBoost (Final) > Random Forest >> Logistic Regression.

4.2 Confusion matrices and error patterns:

Confusion matrices give a more detailed view of how mistakes are spread across classes.

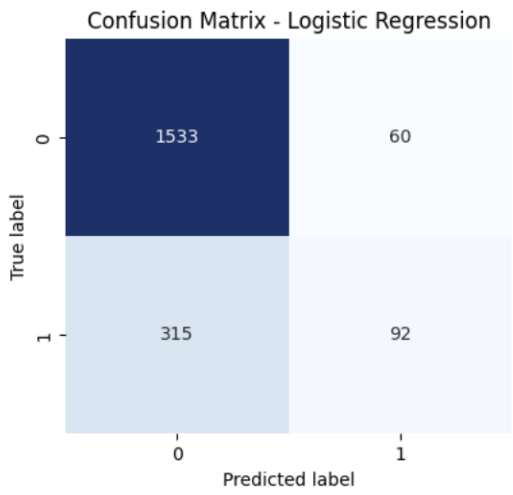

Figure 8: Confusion Matrix - Logistic Regression

The figure 8 represents, Logistic Regression produced 1,533 true negatives, 60 false positives, 315 false negatives and 92 true positives. Most churners (315 out of 407) were wrongly predicted as non-churn. From a business point of view this is a very serious problem, because many customers who are likely to leave would not be chosen for retention actions.

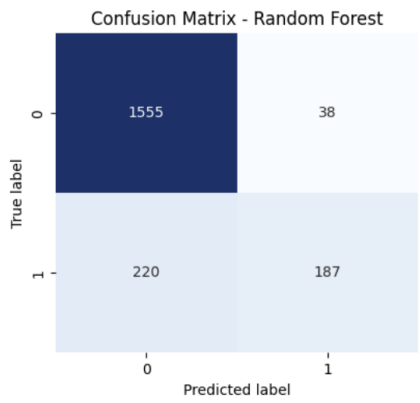

Figure 9: Confusion Matrix - Random Forest

The figure 9 shows, The Random Forest confusion matrix showed 1,555 true negatives, 38 false positives, 220 false negatives and 187 true positives. Compared with Logistic Regression, the number of missed churners fell by almost one third, and false alarms for non-churners also went down. Random Forest therefore gives a better balance between finding churners and avoiding contact with customers.

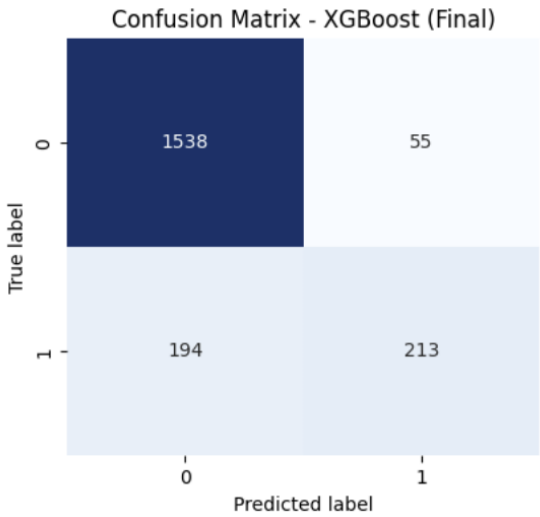

Figure 10: Confusion Matrix - XGBoost (Final)

The figure 10 displays, XGBoost produced 1,538 true negatives, 55 false positives, 194 false negatives and 213 true positives.

Compared with Random Forest, XGBoost found 26 churners at the cost of 17 more false positives. In many banking cases the cost of missing a churner is higher than the cost of calling a non-churner, so higher recall is usually preferred. The confusion matrices support that XGBoost gives the best balance of errors for churn decisions.

4.3 Threshold tuning for XGBoost:

Default predictions use a probability threshold of 0.5, but the best threshold depends on the business goal. To study this, accuracy and recall for churn were calculated for the XGBoost model at thresholds between 0.2 and 0.8.

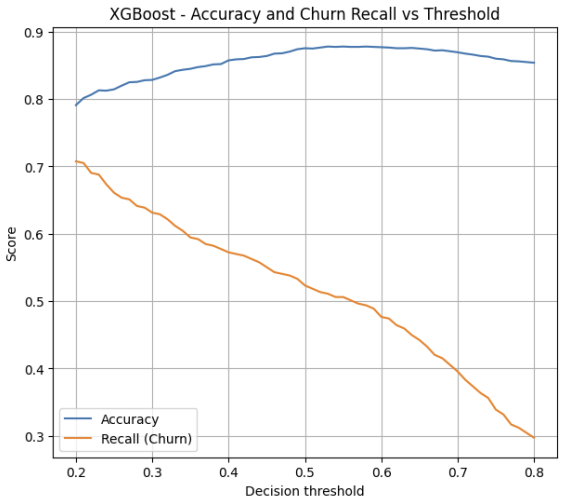

Figure 11: Threshold tune for XGBoost - Accuracy vs Churn Recall vs Threshold

The figure 11 shows that accuracy rises slowly from about 0.79 at threshold 0.2 to a peak of around 0.878 near 0.53, and then starts to fall again. Recall moves in the opposite way, dropping from about 0.71 at very low thresholds to below 0.30 at threshold 0.8. At the threshold that gives the highest accuracy (0.53), churn recall is 0.5111. This means a bit more than half of all churners are found when overall accuracy is at its best.

If the main aim is to find as many churners as possible, a lower threshold can be chosen. For example, in the range 0.40–0.45, recall is above 0.60 while accuracy stays close to 0.86. On the other hand, if the bank wants to limit too many false churn alerts, a slightly higher threshold can be used with only a small drop in recall. This threshold test shows that XGBoost can be tuned to different business cost choices without training the model again.

4.4 Robustness and limitations:

Extra checks with different random seeds for the train–test split showed that, although exact accuracy and ROC-AUC scores changed a little, the order of the three models stayed the same. XGBoost always matched or beat Random Forest, and both ensemble models were clearly better than Logistic Regression. This stable pattern increases trust that the comparison is not driven by only one lucky split.

One remaining limit is the moderate class imbalance and the focus on standard metrics instead of direct cost-sensitive tuning. Methods such as weighted loss functions, oversampling the minority churn

class, or direct optimisation of expected profit could raise recall, but they would also add more complexity and may make the training process harder to explain.

Overall, the findings show that ensemble tree models are more effective than a linear baseline for bank customer churn prediction on this dataset, and that XGBoost in particular gives the best mix of accuracy, ROC-AUC and sensitivity for the churn class.

5. Conclusion and Further Work:

This study looked at bank customer churn prediction using the Kaggle Churn Modelling dataset and three supervised learning models: logistic regression, Random Forest and XGBoost. A clear preprocessing and evaluation pipeline was built with stratified train–test splitting, one-hot encoding for category features and suitable scaling for the linear model. The results showed a clear order in performance. Logistic regression gave a simple baseline but had low recall for churners. Random Forest strongly improved both accuracy and the churn-class F1-score. XGBoost reached the best overall scores, with 87.55% accuracy and ROC-AUC 0.8564, as well as the highest recall and F1-score for the churn class. Threshold-sensitivity analysis also showed how the XGBoost model can be tuned to favour either total accuracy or better detection of at-risk customers.

Future work could grow this project in several ways. Cost-sensitive learning or profit-based evaluation would let the model aim directly at business value instead of general metrics. Extra methods for handling class imbalance, such as focal loss or careful class weights, could be tested to raise churn recall. Temporal checks on time-ordered customer data would test how stable the model is in real deployment. Finally, model-agnostic explanation tools (for example, SHAP) could give more detailed insight into single churn predictions.

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